

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402237
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Wei; Wei et al.

Transverse wave excitation plasma array generators

Abstract

A transverse wave excitation plasma array generator is provided, comprising a plasma array controller and a controlled profile. The plasma array controller includes a plurality of plasma generators, and the plasma array controller is mounted on the controlled profile. The plasma array controller is configured to: control the plurality of plasma generators to generate high-frequency jets under an excitation of high-frequency electricity, obtain spatially distributed high-frequency transverse waves, and generate, based on the high-frequency transverse waves, high-frequency excitation for controlling a second mode of a boundary layer of the controlled profile to promote transition of the boundary layer of the controlled profile, improving a boundary of a frequency domain of plasma flow control. Besides, each of the plurality of plasma generators is connected in series with a plurality of capacitor units and the plurality of inductive coils to form a multi-level array loop. A lower array loop is connected with the inductive coil of a higher array loop to form a loop module. A plurality of the loop modules are connected in series to form a loop module group. A plurality of the loop module groups are connected in parallel to form the plasma array controller.

Inventors: Wei; Wei (Beijing, CN), Guo; Fengling (Beijing, CN), Yang; Hui (Beijing, CN)

Applicant: CHINA ACADEMY OF AEROSPACE AERODYNAMICS (Beijing, CN)

Family ID: 1000008778886

Assignee: CHINA ACADEMY OF AEROSPACE AERODYNAMICS (Beijing, CN)

Appl. No.: 18/434794

Filed: February 06, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250024581 A1	Jan. 16, 2025

Related U.S. Application Data

Publication Classification

Int. Cl.: H05H1/46 (20060101)

U.S. Cl.:

CPC H05H1/466 (20210501); H05H1/4652 (20210501);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2063947	12/1935	Pierce	318/118	G01S 3/805
3932039	12/1975	Frey	356/138	G01D 5/345
5369511	12/1993	Amos	359/107	G02B 19/009
8579835	12/2012	Britva	601/2	A61N 7/00
8726734	12/2013	Lin	73/626	B06B 1/0622
10959703	12/2020	Hollender	N/A	G01N 29/0654
11205410	12/2020	Kim	N/A	G10K 11/172
11255663	12/2021	Binder	N/A	G01S 15/08
11327163	12/2021	Palmeri	N/A	G01S 7/52026
11612381	12/2022	Crocco	600/438	A61B 8/5223
11633175	12/2022	Questa	600/438	A61B 8/461
11906290	12/2023	Binder	N/A	G01B 11/026
2003/0214199	12/2002	Heim	310/309	H04R 5/02
2004/0194547	12/2003	Tozaki	73/579	G01N 29/12
2010/0052686	12/2009	Park	324/332	G01V 11/007
2010/0301398	12/2009	Rothberg	257/253	G01N 27/4145
2011/0213279	12/2010	Britva	601/2	A61N 7/00
2012/0261274	12/2011	Rearick	205/789	G01N 27/4145
2012/0265474	12/2011	Rearick	702/104	G16B 30/00
2012/0302820	12/2011	Carmeli	600/12	A61B 5/0538
2013/0231870	12/2012	Sugnet	702/19	G16B 40/00
2014/0093881	12/2013	Sugnet	435/6.12	H03M 7/30
2014/0113828	12/2013	Gilbert	252/500	G01K 7/006
2015/0249341	12/2014	Kapcia	307/104	H02J 50/12
2015/0303895	12/2014	Ballandras	310/313R	H10N 30/40
2016/0032906	12/2015	Leskosek	60/203.1	F03H 1/0006

2017/0232122	12/2016	Hancock	604/23	A61L 2/14
2018/0296189	12/2017	Hollender	N/A	G01N 29/07
2018/0374466	12/2017	Kim	N/A	G10K 11/162
2019/0041506	12/2018	Palmeri	N/A	G01S 7/52036
2019/0154439	12/2018	Binder	N/A	G01S 15/42
2019/0328364	12/2018	Questa	N/A	A61B 8/485
2019/0350559	12/2018	Bini	N/A	A61B 8/485
2020/0094066	12/2019	Heath	N/A	A61N 5/0616
2021/0022710	12/2020	Crocco	N/A	G01S 15/8915
2021/0359660	12/2020	Tanno	N/A	H03H 9/6483
2022/0104794	12/2021	De Beni	N/A	A61B 8/5253
2022/0128352	12/2021	Binder	N/A	G01S 13/878
2024/0159522	12/2023	Binder	N/A	G01S 15/42
2024/0175678	12/2023	Binder	N/A	G01B 15/00
2025/0024581	12/2024	Wei	N/A	H05H 1/46

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2713939	12/2016	CA	A61N 7/00
102114910	12/2010	CN	N/A
107238481	12/2016	CN	N/A
110131072	12/2018	CN	N/A
110920869	12/2019	CN	N/A
111520352	12/2019	CN	N/A
112594011	12/2020	CN	N/A
114364115	12/2022	CN	N/A
207343	12/2015	IL	A61N 7/00

OTHER PUBLICATIONS

International Search Report in PCT/CN2022/072331 mailed on Jul. 1, 2022, 9 pages. cited by applicant

Written Opinion in PCT/CN2022/072331 mailed on Jul. 1, 2022, 7 pages. cited by applicant

International Search Report in PCT/CN2022/072331 mailed on Jul. 1, 2022, 7 pages. cited by applicant

Written Opinion in PCT/CN2022/072331 mailed on Jul. 1, 2022, 5 pages. cited by applicant

Primary Examiner: Sathiraju; Srinivas

Attorney, Agent or Firm: Porus IP LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation-in-Part of the International Patent Application No. PCT/CN 2022/072331, filed on Jan. 17, 2022, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) The present disclosure relates to the technical field of plasma, and in particular to a transverse wave excitation plasma array generator.

BACKGROUND

(2) The flow state of a boundary layer has a significant influence on the heat flow and friction resistance of an aircraft surface. In high-speed aircrafts, there may be a dynamic coupling phenomenon between a transition state and an aircraft motion/control, impacting the flight stability of the aircraft. Therefore, the control of flow transition is one of the key issues in the development of high-speed aircrafts. There are multiple factors affecting transition, including incoming flow conditions, surface roughness, the structural form of a transition zone, etc. The form of the transition is also categorized as natural transition, forced transition, etc., especially regarding the excitation/control of a second mode (a main frequency being approximately 50 kHz) of the boundary layer.

(3) At present, flow transition is often controlled using passive control structures (e.g., a vortex generator, etc.). However, this approach cannot cover a relatively wide flight envelope of high-speed aircrafts, and can only select limited operating conditions (i.e., an operating state of a device) for flow control. Other flow transition control structures (e.g., blowing and suction systems, diaphragm and piezoelectric synthetic jet generators, etc.) often fail to adequately meet the control needs of the second mode of the boundary layer for high-speed aircrafts due to a frequency, an intensity, a power, an angle, or other factors, resulting in a reduced flow control boundary.

(4) Therefore, it is desirable to provide a transverse wave excitation plasma array generator to improve the boundary of plasma flow control.

SUMMARY

(5) One of the embodiments of the present disclosure provides a transverse wave excitation plasma array generator, comprising a plasma array controller and a controlled profile. The plasma array controller may include a plurality of plasma generators. The plasma array controller may be mounted on the controlled profile. The plasma array controller may be configured to: control the plurality of plasma generators to generate high-frequency jets under an excitation of high-frequency electricity, obtain spatially distributed high-frequency transverse waves, and generate, based on the high-frequency transverse waves, high-frequency excitation for controlling a second mode of a boundary layer of the controlled profile to promote transition of the boundary layer of the controlled profile. The plasma array controller may further include a capacitor and an inductor. The capacitor may include a plurality of capacitor units. The inductor may include a plurality of inductive coils. Each of the plurality of plasma generators may be connected in series with the plurality of capacitor units and the plurality of inductive coils to form a multi-level array loop. An input end of a lower array loop may be connected with an output end of the inductive coil of a higher array loop to form a loop module. A plurality of the loop modules may be connected in series to form a loop module group. A plurality of the loop module groups may be connected in parallel to form the plasma array controller.

(6) In some embodiments, the plurality of the plasma generators may be connected with the capacitor and the inductor to form an inductance-capacitance resonance circuit. The inductance-capacitance resonance circuit may be set at a predetermined loop position.

(7) In some embodiments, each of the plurality of plasma generators may include a cavity and electrodes disposed at two sides of the cavity. The electrodes may generate, based on the high-frequency electricity, the high-frequency jets.

(8) In some embodiments, a material of the cavity may include a ceramic, and a material of the electrodes may include tungsten.

(9) In some embodiments, a ratio of a diameter of each of the electrodes to an inner diameter of the cavity may be within a range of 0.05-0.08.

(10) In some embodiments, a frequency of the high-frequency electricity may be within a range of 1 KHz-3 KHz, and a voltage of the high-frequency electricity may be within a range of 1 KV-10 KV.

(11) In some embodiments, a ratio of a spacing of the plurality of plasma generators in an X-direction to a spacing of the plurality of plasma generators in a Z-direction may be within a range

of 0.8-1.2.

(12) In some embodiments, output characteristics of the plurality of plasma generators may satisfy a predetermined output relationship expressed as:

$$Y=F(\omega,t,\tau,\varphi)$$

wherein Y denotes the output characteristics of the plurality of plasma generators, ω denotes a jet frequency of the plurality of plasma generators, t denotes a time sequence of the plurality of plasma generators generating the high-frequency jets, τ denotes a spatial distribution delay matrix of the transverse wave excitation plasma array generator, and φ denotes a spatial distribution phase matrix of the transverse wave excitation plasma array generator.

(13) In some embodiments, one or more gas pressure sensors may be provided within a predetermined range where the boundary layer of the controlled profile is located. The plasma array controller may be further configured to obtain a gas pressure sequence detected by the one or more gas pressure sensors, and determine, based on a distance between the gas pressure sequence and a predetermined gas pressure sequence, an emission parameter of the plurality of plasma generators. The emission parameter may include an arrangement of the plurality of plasma generators.

(14) In some embodiments, the emission parameter may further include a time delay and a phase of the high-frequency jets output from the plurality of plasma generators. The plasma array controller may be further configured to predict, based on a predetermined high-frequency transverse wave feature, the time delay and the phase of the high-frequency jets output from the plurality of plasma generators through an emission prediction model, the emission prediction model being a machine learning model; and determine the arrangement of the plurality of plasma generators based on the time delay and the phase of the high-frequency jets.

(15) In some embodiments, a count and an arrangement of the one or more gas pressure sensors may be related to a type of transition, and a demand degree of transition of the boundary layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will be further illustrated by way of exemplary embodiments, which will be described in detail with reference to the accompanying drawings. These embodiments are not limiting. In these embodiments, the same numbering indicates the same structure, wherein:

(2) FIG. 1 is a schematic structural diagram illustrating a transverse wave excitation plasma array generator according to some embodiments of the present disclosure;

(3) FIG. 2 is a schematic structural diagram illustrating a circuit of a plasma array controller according to some embodiments of the present disclosure;

(4) FIG. 3 is a schematic structural diagram illustrating a plurality of plasma generators according to some embodiments of the present disclosure;

(5) FIG. 4 is a schematic structural diagram illustrating a transverse wave excitation plasma array generator according to some embodiments of the present disclosure;

(6) FIG. 5 is a schematic diagram illustrating a working principle of a transverse wave excitation plasma array generator according to some embodiments of the present disclosure;

(7) FIG. 6 is a schematic diagram illustrating a scenario of applying to a conical standard model according to some embodiments of the present disclosure; and

(8) FIG. 7 is a schematic diagram illustrating a process of determining an arrangement of a plurality of plasma generators according to some embodiments of the present disclosure.

REFERENCE SIGNS

(9) **10**: Transverse wave excitation plasma array generator, **100**: plasma array controller, **110**: plasma generator, **111**: cavity, **112**: electrode, **120**: capacitor, **130**: inductive coil, **140**: loop module,

141: array loop, **114-1:** first array loop, **114-2:** second array loop, **150:** loop module, **200:** controlled profile, **710:** gas pressure sequence, **720:** predetermined gas pressure sequence, **730:** arrangement, **740:** high-frequency transverse wave feature, **750:** emission prediction model, and **760:** time delay and phase of high-frequency jets.

DETAILED DESCRIPTION

(10) In order to more clearly illustrate the technical solutions of the embodiments of the present disclosure, the following briefly introduces the drawings that need to be used in the description of the embodiments. Apparently, the accompanying drawings in the following description are only some examples or embodiments of the present disclosure, and those skilled in the art can also apply the present disclosure to other similar scenarios according to the drawings without creative efforts. Unless obviously obtained from the context or the context illustrates otherwise, the same numeral in the drawings refers to the same structure or operation.

(11) It should be understood that “system”, “device”, “unit” and/or “module” as used herein is a method for distinguishing different components, elements, parts, portions or assemblies of different levels. However, the words may be replaced by other expressions if other words can achieve the same purpose.

(12) As indicated in the disclosure and claims, the terms “a”, “an” and/or “the” are not specific to the singular form and may include the plural form unless the context clearly indicates an exception. Generally speaking, the terms “comprising” and “including” only suggest the inclusion of clearly identified steps and elements, and these steps and elements do not constitute an exclusive list, and the method or device may also contain other steps or elements.

(13) The flowchart is used in the present disclosure to illustrate the operations performed by the system according to the embodiments of the present disclosure. It should be understood that the preceding or following operations are not necessarily performed in the exact order. Instead, various steps may be processed in reverse order or simultaneously. Meanwhile, other operations may be added to these procedures, or a certain step or steps may be removed from these procedures.

(14) Regarding how to control flow transition, passive control structures (e.g., vortex generators, etc.) or other flow transition control structures (e.g., a blowing/suction system, diaphragm and piezoelectric synthetic jet generators, etc.) may be adopted in the prior art. These structures may transition a fluid from a laminar state to a turbulent state when encountering a head-on fluid, thereby facilitating the process of flow transition.

(15) Merely by way of example, the vortex generator in the prior art is generally arranged on a surface of a wing or a flap to generate a flow vortex, enhancing mixing of main flow and boundary layer flow to achieve the purpose of suppressing separation. However, vortex generator may not improve performance under different flight operating conditions. The blowing/suction system in the prior art may inject energy into the boundary layer by adding an external air source, but an equipment size and power consumption of the blowing/suction system are too large. The diaphragm and piezoelectric synthetic jet generators in the prior art generate low-energy high-frequency jets with relatively low jet velocities. A flow transition effect produced by the diaphragm and piezoelectric synthetic jet generators is still insufficient to meet practical standards, which leads to few applications in high-speed flows. A piston synthetic jet generator in the prior art may achieve a relatively high energy output. Due to a structural limitation of the piston synthetic jet generator, a jet frequency of the piston synthetic jet generator cannot reach a relatively high level.

(16) To sum up, the control requirements of a high-speed aircraft for a second mode of a boundary layer are difficult to satisfy in the prior art with respect to factors such as operating condition coverage, frequency, intensity, power, etc.

(17) The present disclosure provides a transverse wave excitation plasma array generator. The transverse wave excitation plasma array generator may promote forced transition by generating high-frequency transverse waves and controlling the high-frequency transverse waves to resonate with the second mode of the boundary layer. In addition, the transverse wave excitation plasma

array generator may also adjust the high-frequency transverse waves in real time based on a flight state of the aircraft to satisfy the control requirements of the second mode of the boundary layer. Furthermore, the high-frequency transverse waves generated by the transverse wave excitation plasma array generator may be superimposed and combined along a flow direction, resulting in frequency modulation and intensity adjustment of the flow direction, thereby widening the boundary of flow control.

(18) It should be understood that the application scenarios of the transverse wave excitation plasma array generator of the present disclosure are only some examples or embodiments of the present disclosure, and those having ordinary skills in the art, without creative efforts, may apply the present disclosure to other similar scenarios in accordance with these drawings.

(19) The transverse wave excitation plasma array generator will be illustrated in detail according to the embodiments of the present disclosure with reference to FIGS. 1-7. It should be noted that the following embodiments are intended only to explain the present disclosure and do not constitute a limitation of the present disclosure.

(20) FIG. 1 is a schematic structural diagram illustrating a transverse wave excitation plasma array generator according to some embodiments of the present disclosure. In some embodiments, as illustrated in FIG. 1, a transverse wave excitation plasma array generator **10** may include a plasma array controller **100** and a controlled profile **200**. The plasma array controller **100** may be mounted on the controlled profile **200**.

(21) The plasma array controller **100** refers to an electronic device that performs transition control. In some embodiments, the plasma array controller **100** may be configured to control a plurality of plasma generators **110** to generate high-frequency jets under an excitation of high-frequency electricity to obtain spatially distributed high-frequency transverse waves. The plasma array controller **100** may generate a high-frequency excitation for controlling a second mode of a boundary layer of a controlled profile based on the high-frequency transverse waves to promote transition of the boundary layer of the controlled profile **200**.

(22) The high-frequency electricity refers to a power supply provided to the plasma generator **110** for operation. In some embodiments, the plurality of plasma generators **110** may generate the corresponding high-frequency jets based on the high-frequency electricity. Correspondingly, the plurality of plasma generators **110** may be arranged on the controlled profile **200**. An arrangement of the plurality of plasma generators **110** may cause the high-frequency jets generated by the plurality of the plasma generators **110** to combine with each other to form the high-frequency transverse waves. More descriptions regarding the plurality of plasma generators **110** may be found in FIG. 2 and related descriptions thereof. More descriptions regarding the arrangement may be found in FIG. 7 and related descriptions thereof.

(23) In some embodiments, a frequency of the high-frequency electricity may be within a predetermined frequency range, and a voltage of the high-frequency electricity may be within a predetermined voltage range. In some embodiments, the predetermined frequency range and the predetermined voltage range may be set based on manual historical experience or a demand of transition. For example, when an aircraft needs to accelerate, and the demand of transition is to accelerate a transition speed, the transverse wave excitation plasma array generator **10** may set a relatively high predetermined frequency range of the frequency and a relatively high predetermined voltage range of the voltage, so that the plurality of plasma generators **110** may generate corresponding high-frequency transverse waves to accelerate the transition of the boundary layer. Merely by way of example, the predetermined frequency range may be within a range of 1 kHz-3 kHz, and the predetermined voltage range may be within a range of 1 kV-10 kV.

(24) In the present disclosure, by designing the frequency of the high-frequency electricity to be within a range of 1 kHz-3 kHz and the voltage of the high-frequency electricity to be within a range of 1 kV-10 kV, the plurality of plasma generators **110** may be controlled to generate a high-frequency transverse flow, thereby achieving frequency modulation and intensity adjustment of the

high-frequency transverse flow in a flow direction, and widening the boundary of flow control.

(25) The high-frequency jets refer to a flow of fluid carrying ions in a high-frequency range. In some embodiments, the plurality of plasma generators **110** may inject an energy into atoms and molecules under the high-frequency electricity, converting the atoms and the molecules into ions to generating the high-frequency jets. In some embodiments, a frequency of the high-frequency jets may be related to the frequency of the high-frequency electricity. For example, the higher the frequency of the high-frequency electricity, the higher the frequency of the generated high-frequency jets. In some embodiments, the frequency of the high-frequency jets may be set within a frequency range in various ways based on manual experience or control requirements. Merely by way of example, the frequency of the high-frequency jets may be within a predetermined jet frequency range. The predetermined jet frequency range may be within a range of 1 kHz-3 kHz.

(26) The high-frequency transverse waves refer to electromagnetic waves in the high-frequency range where a vibration direction of the ions is perpendicular to an advance direction of the ions. When the high-frequency transverse waves propagate in the flow field, the high-frequency transverse waves interact with a gas in a flow field and generate a certain perturbation. When the high-frequency transverse waves interact with gas molecules, the high-frequency transverse waves generate a certain amount of momentum transfer and heat transfer, which alters kinetic energy and thermodynamic states of the gas molecules, causing the gas molecules to move and vibrate. The movement and the vibration may further perturb surrounding gas molecules, causing a chain reaction, and ultimately causing a perturbation in the flow field to achieve transition.

(27) The transition refers to a process of a fluid transitioning from a laminar flow state to a turbulent flow state. In fluid dynamics, a motion state of the fluid may be classified into the laminar flow state and the turbulent flow state. In the laminar flow state, the fluid in a pipe or another container is orderly arranged, with a flow velocity uniformly distributed along a flow channel, presenting laminar flow characteristics. In the turbulent flow state, the motion state of the fluid becomes complex and disordered, with an irregular distribution and magnitude of the flow velocity throughout the flow channel, presenting turbulent flow characteristics.

(28) The boundary layer refers to a fluid boundary layer that attaches to surface of a solid when the fluid flows along the surface of the solid or when the solid moves in the fluid. For example, taking the air as an example, when the air flows over an object, a surface of the object is not perfectly smooth, and the air has viscosity, which causes a layer of air close to the surface of the object to encounter resistance during flowing, causing the flow velocity to reduce to zero and causing the flow velocity of an upper layer of air to reduce. As a result, a thin layer of air, referred to as the boundary layer, where the flow velocity gradually increases along a normal direction to the surface of the object, may be generated close to the surface of the object.

(29) The second mode refers to an inherent vibration characteristic of the boundary layer with a predetermined inherent frequency. In some embodiments, the predetermined inherent frequency may be determined based on a parameter and a morphology of the boundary layer. For example, the predetermined inherent frequency may include a main frequency of approximately 50 kHz. In some embodiments, when the second mode of the boundary layer is subjected to the high-frequency excitation, the boundary layer of the controlled profile **200** may transition from the laminar flow state to the turbulent flow state, thereby actively achieving the forced transition of the boundary layer.

(30) The high-frequency excitation refers to an excitation process of providing an energy in the high-frequency range to a substance and causing the substance to undergo a transformation. In some embodiments, the high-frequency transverse waves may interact with the fluid in the boundary layer, so that the second mode of the boundary layer may be subjected to the energy provided by the high-frequency transverse waves, and thus the fluid in the boundary layer may transition from the laminar flow state to the turbulent flow state, achieving the transition of the boundary layer.

- (31) In some embodiments, the plasma array controller may include the plurality of plasma generators **110** to generate the spatially distributed high-frequency transverse waves.
- (32) The plurality of plasma generators **110** refer to electronic devices that generate a large number of positive or negative ions. In some embodiments, each of the plurality of plasma generators **110** may generate the high-frequency jets on the boundary layer of the controlled profile **200** under the high-frequency electricity. Correspondingly, a combination of the plurality of plasma generators **110** may generate the spatially distributed high-frequency transverse waves on the boundary layer of the controlled profile **200**.
- (33) The spatial distribution refers to a distribution state of the plurality of plasma generators **110** within a space of the boundary layer of the controlled profile **200**. In some embodiments, the spatial distribution may include one or more distributions of a spaced distribution, a tiled distribution, a layered distribution, or the like. The spaced distribution means that the plurality of the plasma generators **110** are arranged apart at a predetermined distance. The tiled distribution means that the plurality of the plasma generators **110** are distributed on a plane where the controlled profile **200** is located. The layered distribution means that at least two plasma generators of the plurality of the plasma generators **110** are present in an overlapping distribution within the space of the boundary layer of the controlled profile **200**.
- (34) FIG. 2 is a schematic structural diagram illustrating a circuit of a plasma array controller according to some embodiments of the present disclosure. In some embodiments, a capacitor may include a plurality of capacitors **120**, and an inductor may include a plurality of inductive coils **130**. Each of the plurality of plasma generators **110** may be connected in series with the plurality of capacitor units **120** and the plurality of inductive coils **130** to form a multi-level array loop **141**. An input end of a lower array loop **141-2** may be connected with an output end of the inductive coil **130** of a higher array loop **141-1** to form a loop module **140**. A plurality of the loop modules **140** may be connected in series to form a loop module group **150**. A plurality of the loop module groups **150** may be connected in parallel to form the plasma array controller **100**.
- (35) As illustrated in FIG. 2, each of the multi-level array loop **141** may include an inductive coil L.sub.z, x, a capacitor unit C.sub.z, x, and a plasma generator Jet.sub.z, x connected in series in sequence. The plasma generator Jet.sub.z, x may be controlled by the capacitor unit C.sub.z, x and the inductive coil L.sub.z, x to output corresponding high-frequency jets.
- (36) As illustrated in FIG. 2, the input end of the lower array loop **141-2** may be connected with the output end (i.e., a connection point between the capacitor C.sub.z, x and the inductive coil L.sub.z, x) of the inductive coil **130** of the higher array loop **141-1**, so that the lower array loop **141-2** and the higher array loop **141-1** may form the loop module group **150**.
- (37) As illustrated in FIG. 2, in a series connection of the plurality of loop modules **140**, an input end of the inductive coil L.sub.z, x in a first array loop **141** of a next loop module **140** may be connected with an output end of the plasma generator **110** of a previous loop module **140** to form the loop module group **150** in the series connection.
- (38) As illustrated in FIG. 2, in a parallel connection of the plurality of loop module groups **150**, input ends of the plurality of inductive coil L.sub.z, x in the first array loop **141** of each of the plurality of loop module groups **150** may be connected, and output ends of the plurality of plasma generators in a last array loop **141** of each of the plurality of loop module groups **150** may be connected. The connected input ends and the connected output ends may receive an excitation of the high-frequency electricity, respectively, to form the plasma array controller **100**.
- (39) In some embodiments of the present disclosure, the multi-level array loop **141**, the plurality of loop modules **140**, and the plurality of loop module groups **150** may be provided to control each of the plurality of plasma generators **110** and separately drive the corresponding plasma generators **110** to output the high-frequency transverse flow, thereby improving the adjustability of the high-frequency transverse flow, and expanding the boundary of flow control.
- (40) FIG. 3 is a schematic structural diagram illustrating a plurality of plasma generators **110**

according to some embodiments of the present disclosure. In some embodiments, as illustrated in FIG. 3, each of the plurality of plasma generators **110** (e.g., the plasma generator Jet.sub.z, x illustrated in FIG. 3) may include a cavity **111** and electrodes **112** disposed at two sides of the cavity **111**. The electrodes **112** may generate, based on high-frequency electricity, high-frequency jets.

(41) In some embodiments, the cavity **111** may provide a space required for generating the high-frequency jets. When an electric field is applied to the cavity **111**, atoms and molecules within the cavity **111** are energized to be converted into ions. Correspondingly, an interior of the cavity **111** may form a plasma region. In some embodiments, the cavity **111** may also be provided with through holes to allow the ions to be ejected from the cavity **111**, thereby generating the high-frequency jets.

(42) In some embodiments, the electrodes **112** may apply the electric field to the cavity **111**, under the high-frequency electricity, injecting an energy into the atoms and the molecules inside the cavity **111** to convert the atoms and the molecules into the ions, thereby generating the high-frequency jets. In some embodiments, the plasma array controller **100** may adjust the high-frequency electricity of the plurality of plasma generators **110** based on a flight state of an aircraft, thereby adjusting the high-frequency transverse waves in real time, and satisfying control requirements of a second mode of a boundary layer. For example, when the flight state is acceleration, the plasma array controller **100** may increase a voltage of the high-frequency electricity to generate relatively large high-frequency transverse waves, causing the boundary layer to quickly transition to avoid fluid impedance and facilitate acceleration of the aircraft.

(43) In some embodiments, a material of the cavity **111** may include one or more insulating materials such as a ceramic, rubber, or plastic. In some embodiments, a material of the electrodes **112** may include one or more electrically conductive materials such as tungsten, copper, or silver. In some embodiments, a material of the cavity **111** may include a ceramic, and a material of the electrodes **112** may include tungsten.

(44) In the embodiments of the present disclosure, the cavity **111**, which is made of the ceramic, enhances the insulation of an environment where ions are generated, and the electrodes **112**, which is made of the tungsten, ensures stable supply of the high-frequency electricity. This design can control steady generation of the high-frequency jets by the plurality of plasma generators **110** to satisfy the requirements of providing a high-frequency excitation to the second mode of the boundary layer, thereby achieving transition control.

(45) In some embodiments, a ratio of a diameter of each of the electrodes **112** to an inner diameter of the cavity **111** may be within a predetermined diameter ratio range. In some embodiments, the predetermined diameter ratio range may be set based on manual experience or a demand of transition. For example, the ratio of the diameter of each of the electrodes **112** to the inner diameter of the cavity **111** may be within a range of 0.05-0.08. For example, the ratio of the diameter of each of the electrodes **112** to the inner diameter of the cavity **111** may be 0.05, 0.06, 0.08, or the like.

(46) In the embodiments of the present disclosure, by setting the ratio of the diameter of each of the electrodes **112** to the inner diameter of the cavity **111** to be within the range of 0.05-0.08, the plurality of plasma generators may be controlled to generate the high-frequency jets that satisfy the demand of transition, thereby generating the high-frequency transverse waves to excite the second mode of the boundary layer and promote forced transition.

(47) In the embodiments of the present disclosure, through the cooperation of the electrodes and the cavity, the plasma array controller **100** may adjust the high-frequency electricity applied to the plurality of plasma generator based on the flight state of the aircraft, thereby adjusting the high-frequency transverse waves in real time to satisfy the control requirements of the second mode of the boundary layer.

(48) The controlled profile **200** refers to a surface of a device where the boundary layer is controlled, such as a two-dimensional inlet duct profile, a binary retracting and expanding nozzle

runner profile, and other aircraft profiles. In some embodiments, the controlled profile **200** may be provided with the plasma array controller **100**. The plasma array controller **100** may generate spatially distributed high-frequency transverse waves and cause the high-frequency transverse waves to resonate with the second mode of the boundary layer of the controlled profile **200** to promote the forced transition of the boundary layer of the controlled profile **200**.

(49) When a frequency of the high-frequency transverse waves is close to or equal to an inherent frequency of the second mode of the boundary layer of the controlled profiles **200**, the high-frequency transverse waves and the second mode of the boundary layer of the controlled profiles **200** may resonate, thereby providing the high-frequency excitation for the second mode of the boundary layer.

(50) FIG. **4** is a schematic structural diagram illustrating the transverse wave excitation plasma array generator **10** according to some embodiments of the present disclosure. In some embodiments, as illustrated in FIG. **4**, the plasma array controller **100** may be mounted on the controlled profiles **200**. The plurality of the plasma generators **110** may be arranged in the plasma array controller **100** in a predetermined arrangement. In some embodiments, the arrangement may include parameters such as a position, a spacing, and a count of the plurality of the plasma generators **110**. More descriptions regarding the arrangement may be found in FIG. **7** and related descriptions thereof.

(51) For example, an X-direction as illustrated in FIG. **4** represents a transverse direction of the plasma array controller **100**, and a Z-direction as illustrated in FIG. **4** represents a longitudinal direction of the plasma array controller **100**. Nd plasma generators **110** may be provided in the X-direction of the plasma array controller **100**, and Ne plasma generators **110** may be provided in the Z-direction of the plasma array controller **100**.

(52) As illustrated in FIG. **4**, as illustrated in FIG. **4**, a V-direction represents a direction of motion of high-frequency transverse waves generated by the plurality of plasma generators **110**, and a length L represents a transmission length of the high-frequency transverse waves generated by the plurality of plasma generators **110**. In some embodiments, position information of each of the plurality of plasma generators **110** may be represented using coordinates. As illustrated in FIG. **4**, coordinate positions of the plurality of plasma generators **110** may be represented as (z, x).

(53) In some embodiments, in the plasma array controller **100**, a ratio of a spacing of the plurality of the plasma generators **110** in the X-direction to a spacing of the plurality of the plasma generators **110** in the Z-direction may be within a predetermined spacing range. As illustrated in FIG. **4**, the spacing of the plurality of the plasma generators **110** in the transverse direction (i.e., the X-direction) may be denoted as d, and the spacing of the plurality of the plasma generators **110** in the longitudinal direction (i.e., the Z-direction) may be denoted as e, then the ratio of the spacing of the plurality of the plasma generators **110** in the X-direction to the spacing of the plurality of the plasma generators **110** in the Z-direction may be denoted as d/e.

(54) In some embodiments, the predetermined spacing range may be determined based on manual experience or the demand of transition. For example, the ratio d/e of the spacing d of the plurality of the plasma generators **110** in the transverse direction (i.e., the X-direction) to the spacing e of the plurality of the plasma generators **110** in the longitudinal direction (i.e., the Z-direction) may be within a range of 0.8-1.2. For example, the ratio d/e of the spacing d of the plurality of the plasma generators **110** in the transverse direction (i.e., the X-direction) to the spacing e of the plurality of the plasma generators **110** in the longitudinal direction (i.e., the Z-direction) may be 0.8, 1.0, 1.1, 1.2, or the like.

(55) In some embodiments of the present disclosure, by designing the spacing of the plurality of the plasma generators **110** to be within the range of 0.8-1.2, the plurality of plasmas may emit the corresponding spatially distributed the high-frequency transverse waves, thereby achieving stable transition control.

(56) In some embodiments, a ratio of a count of the plurality of the plasma generators distributed in

the X-direction to a count of the plurality of the plasma generators distributed in the Z-direction may be within a predetermined count range. As illustrated in FIG. 4, the count of the plurality of plasma generators **110** distributed in the X-direction may be denoted as Nd, and the count of the plurality of plasma generators **110** distributed in the Z-direction may be denoted as Ne. The ratio of the count of the plurality of the plasma generators distributed in the X-direction to the count of the plurality of the plasma generators distributed in the Z-direction may be denoted as Nd/Ne.

(57) In some embodiments, the predetermined count range may be determined based on manual experience or the demand of transition. For example, the ratio Nd/Ne of the count of the plurality of the plasma generators distributed in the X-direction to the count of the plurality of the plasma generators distributed in the Z-direction may be within a range of 0.8-1.2. For example, the ratio Nd/Ne of the count of the plurality of the plasma generators distributed in the X-direction to the count of the plurality of the plasma generators distributed in the Z-direction may be 0.8, 0.9, 1.2, or the like.

(58) In some embodiments of the present disclosure, by designing the count of the plurality of the plasma generators **110** in the longitudinal and the transverse directions to be within the range of 0.8-1.2, the plurality of the plasma generators may be distributed on the controlled profile **200** in a tiled manner, thereby emitting the corresponding spatially distributed high-frequency transverse waves to achieve stable transition control.

(59) In some embodiments, output characteristics of the plurality of plasma generators **110** may satisfy a predetermined output relationship. The predetermined output relationship may be expressed by equation (1):

$$(60) \quad Y = F(\omega, t, \tau, \varphi) \quad (1)$$

(61) Wherein Y denotes the output characteristics of the plurality of plasma generators, ω denotes a jet frequency of the plurality of plasma generators, t denotes a time sequence of the plurality of plasma generators generating the high-frequency jets, τ denotes a spatial distribution delay matrix of the transverse wave excitation plasma array generator **10**, and φ denotes a spatial distribution phase matrix of the transverse wave excitation plasma array generator **10**.

(62) The output characteristics of the plurality of plasma generators **110** refer to a characteristic relationship between a current and a voltage of plasma generation. The jet frequency refers to an angular frequency of the plurality of plasma generators **110** for outputting the high-frequency transverse waves. The time sequence refers to a combination of a plurality of time points at which the plurality of plasma generators **110** are required to generate the high-frequency jets. The spatially distributed delay matrix refers to a combination of output delay vectors between the plurality of the plasma generators **110**. The spatial distribution phase matrix refers to a combination of output phase vectors between the plurality of the plasma generators **110**.

(63) For example, as illustrated in FIG. 4, output characteristics $Y_{z,x}$ of a plasma generator (z, x) may be represented by equation (2):

$$(64) \quad Y_{z,x} = \sin(\omega_{z,x}(t + \tau(z,x) + \varphi(z,x))) \quad (2)$$

(65) Wherein $Y_{z,x}$ denotes the output characteristics of the plasma generator (z,x), (z,x) denotes the coordinate position of the plasma generator, $\omega_{z,x}$ denotes the jet frequency of the plasma generator (z,x), t denotes the time sequence of the plasma generator (z,x) generating the high-frequency jets, $\tau(z,x)$ denotes the spatial distribution delay matrix of the transverse wave excitation plasma array generator **10**, and $\varphi(z,x)$ denotes the spatial distribution phase matrix of the transverse wave excitation plasma array generator **10**.

(66) In some embodiments, under the excitation of the high-frequency electricity, the plasma array controller **100** may control the plurality of plasma generators **110** to generate the high-frequency jets to obtain the spatially distributed high-frequency transverse waves, and generate the high-frequency excitation for controlling the second mode of the boundary layer of the controlled profile **200** based on the high-frequency transverse waves, thereby promoting the transition of the

boundary layer of the controlled profile **200**.

(67) FIG. 5 is a schematic diagram illustrating a working principle of a transverse wave excitation plasma array generator according to some embodiments of the present disclosure. As illustrated in FIG. 5, output characteristics of the N plasma generators **110** distributed in an X-direction may be $Y_{sub.1} = \sin(\omega t + 1 \cdot \varphi)$, $Y_{sub.2} = \sin(\omega t + 2 \cdot \varphi)$, . . . $Y_{sub.N} = \sin(\omega t + N \cdot \varphi)$, respectively. That is to say, high-frequency jets emitted by the plurality of the plasma generators **110** may be combined to form high-frequency transverse waves, so that the high-frequency transverse waves may resonate with a second mode of a boundary layer of the controlled profile **200**, thereby promoting transition of the boundary layer of the controlled profile **200**. More descriptions regarding the resonance and the transition may be found in FIG. 1 and the related descriptions thereof.

(68) It should be noted that in some embodiments, the high-frequency jets emitted by the plurality of plasma generators **110** may also be superimposed in a flow direction. As illustrated in FIG. 5, the plurality of the plasma generators **110** distributed in a Z-direction may be superimposed in a flow direction V to increase an intensity of the high-frequency transverse waves.

(69) In some embodiments of the present disclosure, by designing the output characteristics of the plurality of plasma generators **110**, each of the plurality of plasma generators **110** may emit the high-frequency jets, a combination of the plurality of the plasma generators **110** may output a spatially distributed high-frequency transverse flow, thereby achieving forced transition of the boundary layer.

(70) FIG. 6 is a schematic diagram illustrating a scenario of applying to a conical target model according to some embodiments of the present disclosure. In a velocity profile of a boundary layer illustrated in FIG. 6, a curve 1 illustrates a velocity of fluid in the boundary layer after the transverse wave excitation plasma array generator **10** is turned off, and a curve 2 illustrates a velocity of the fluid in the boundary layer after the transverse wave excitation plasma array generator **10** is turned on. That is, the transverse wave excitation plasma array generator **10** may generate high-frequency transverse waves in an opposite direction of a high-velocity incoming flow to transition the fluid in the boundary layer into a turbulent flow, resulting in a decrease in a fluid velocity.

(71) It should be noted that the transverse wave excitation plasma array generator **10** in some embodiments of the present disclosure is particularly suitable for controlling boundary layer flowing in a hypersonic inlet, separated flowing on a high-speed wing surface, shockwave control in a compression corner, or the like. As an example of controlling boundary layer flowing in the hypersonic inlet illustrated in FIG. 6, the transverse wave excitation plasma array generator **10** may be arranged in an inlet channel, so that periodic reverse jet generated by the transverse wave excitation plasma array generator **10** may interact with the high-velocity incoming flow. The interaction may enhance mixing of the boundary layer/mainstream, causing the incoming fluid of the air inlet to transition to the turbulent flow, and effectively suppressing flow separation in the compression corner and a reflection zone of lip shockwaves, thereby improving the performance of the air inlet.

(72) In some embodiments, the plurality of the plasma generators **110** may be connected with a capacitor and an inductor to form an inductance-capacitance resonance circuit. The inductance-capacitance resonance circuit may be set at a predetermined loop position.

(73) In some embodiments, the inductance-capacitance resonance circuit may be configured to drive the plurality of the plasma generators **110** to operate, modulating output characteristics of the plurality of the plasma generators **110**. For example, the inductance-capacitance resonance circuit may control the plurality of plasma generators **110** to output corresponding high-frequency jets through charging and discharging of the capacitor and the inductor.

(74) The predetermined loop position refers to a position of a loop under an excitation by high-frequency electricity. In some embodiments, the predetermined loop position may be set according to a demand of transition. For example, a large number of the plasma generators **110** are required,

the plurality of the plasma generators **110** may be set to be separately located on a loop array, so that the plurality of the plasma generators **110** may generate the high-frequency jets with the same or similar voltage under a same voltage excitation.

(75) In the embodiments of the present disclosure, by setting the inductance-capacitance resonance circuit to drive the plurality of plasma generators **110**, each of the plurality of plasma generators **110** may output the corresponding high-frequency jets, and the plurality of the plasma generators **110** may generate a corresponding high-frequency transverse flow, thereby realizing frequency modulation and intensity adjustment of the high-frequency transverse flow in the flow direction, and broadening the boundary of flow control.

(76) In some embodiments, one or more gas pressure sensors may also be provided within a predetermined range where the boundary layer of the controlled profile **200** is located.

(77) In some embodiments, the one or more gas pressure sensors may be configured to detect a gas pressure of the boundary layer of the controlled profile **200**. The gas pressure may reflect a flow state of the fluid. For example, the lower the gas pressure, the more likely the flow state of the fluid is a laminar flow state. In some embodiments, the predetermined range may be set based on control requirements of the controlled profile **200**. For example, when the flow state of the fluid within a pipeline is required to be controlled, the one or more gas pressure sensors may be set at a circumferential wall of the pipeline.

(78) In some embodiments, the one or more gas pressure sensors may detect a fluid pressure of the boundary layer to obtain a gas pressure sequence of the boundary layer. The gas pressure sequence may be a combination of a time point and the fluid pressure.

(79) In some embodiments, the gas pressure sequence may be in various forms, such as an average of gas pressures obtained by the one or more gas pressure sensors, or the gas pressure sequence obtained by the one or more gas pressure sensors, or a gas pressure characteristic sequence of the one or more gas pressure sensors. The gas pressure characteristic sequence may include an absolute value of pressures detected by the one or more gas pressure sensors, a change rate, a distribution uniformity, a change stability, or the like. The absolute value of the pressures refers to a value of a pressure applied by a gas to the one or more gas pressure sensors as detected by the one or more gas pressure sensors at a time point.

(80) In some embodiments, the gas pressure sequence may reflect an actual gas flow characteristic, such as a laminar flow characteristic, a turbulent flow characteristic, a flow around characteristic, or the like. The flow around characteristic may be reflected in a way that the fluid forms a rotating flow region around an object. More descriptions regarding the laminar and turbulent flow characteristics may be found in FIG. 1 and the related descriptions thereof.

(81) In some embodiments, a count and an arrangement of the one or more gas pressure sensors may be related to a type of transition and a demand degree of transition of the boundary layer.

(82) In some embodiments, the arrangement may include one or more arrangements such as a flat distribution, a vertical distribution, or the like. For example, the one or more gas pressure sensors may be distributed vertically on the boundary layer to gas pressures at different heights of the boundary layer.

(83) The type of transition refers to a type of range of fluid transition. In some embodiments, the type of transition may include one or more types such as local transition, full transition, or the like.

(84) In some embodiments, the demand degree of transition reflects a degree to which the fluid transition is required. For example, the more the gases requiring the fluid transition as for the demand degree of transition, the more the finer gas flow characteristic data required. Accordingly, a relatively large number of gas pressure sensors may be provided. As another example, if the demand degree of transition is low, a relatively small number of the gas pressure sensors may be provided near the attachment layer, thereby saving cost while satisfying the demand of transition.

(85) In some embodiments, the plasma array controller **100** may obtain the type of transition and the demand degree of transition of the boundary layer through various ways, such as manual input,

historical transition data, or the like.

(86) In some embodiments, the type of transition and the demand degree of transition of the boundary layer may affect the count and the arrangement of the one or more gas pressure sensors. For example, if the demand degree of transition is low, a relatively small number of the gas pressure sensors may be provided near the boundary layer to reduce resources consumed for transition detection. As another example, if the demand degree of transition is high, the fluid may be mostly required to be converted from the laminar flow to turbulent flow, and a relatively large number of gas pressure sensors may be provided to obtain more and finer gas flow characteristic data.

(87) In some embodiments, the plasma array controller **100** may determine the count and the arrangement of the corresponding gas pressure sensors using a gas pressure sensor array database based on the type of transition and the demand degree of transition of the boundary layer.

(88) In some embodiments, the gas pressure sensor array database may be determined based on experimental data. For example, the plasma array controller **100** may establish the gas pressure sensor array database by recording a count and an arrangement pattern of different gas pressure sensors corresponding to different levels of transition by cumulatively calculating geometric mean points over a number of experiments (e.g., 20 times), thereby creating the gas pressure sensor array database.

(89) The level of transition level may be related to the type of transition and the demand degree of transition. For example, the plasma array controller **100** may rank the transition based on the type of transition and the demand degree of transition. For example, the closer the type of transition is to the full transition and the demand degree of transition is in a relatively high predetermined demand range, the transition may be classified as level 1.

(90) Correspondingly, the plasma array controller **100** may perform experiments (e.g., 10 experiments, etc.) with a same level (e.g., level 1) of transition in the database, and geometrically average (e.g., Euclidean averaging) corresponding arrangement patterns of the one or more gas pressure sensors, to obtain the arrangement and the count of the one or more gas pressure sensors corresponding to transition with level 1.

(91) In the embodiments of the present disclosure, by setting the count and the arrangement of the one or more gas pressure sensors, the detected gas pressure may be more accurate, making it easier to control subsequent transition.

(92) In some embodiments, as illustrated in FIG. 7, the plasma array controller **100** may obtain a gas pressure sequence **710** detected by one or more gas pressure sensors, and determine, based on a distance between the gas pressure sequence **710** and a predetermined gas pressure sequence **720**, an emission parameter of the plurality of plasma generators **110**. The emission parameter includes an arrangement **730** of the plurality of plasma generators **110**.

(93) The predetermined gas pressure sequence **720** refers to the required gas pressure sequence **710** for reflecting a predetermined gas flow characteristic. In some embodiments, the predetermined gas pressure sequence **720** may also be expressed as a predetermined gas pressure characteristic sequence.

(94) In some embodiments, the predetermined gas pressure sequence **720** may be set based on manual experience or the demand of transition. For example, similar transition state data of the boundary layer may be found in historical transition control data based on a required demand of transition, and the historical gas pressure sequence **710** in the corresponding historical transition control data may be used as the current predetermined gas pressure sequence **720**.

(95) In some embodiments, the distance between the gas pressure sequence **710** and the predetermined gas pressure sequence **720** may be a Euclidean distance. The Euclidean distance reflects a similarity between the gas pressure sequence **710** and the predetermined gas pressure sequence **720**. The larger the Euclidean distance, the lower the similarity, which indicates that the current arrangement **730** of the plurality of plasma generators **110** may have difficulty generating

the predetermined gas flow characteristic, and more modifications may be needed for the current arrangement **730**.

(96) In some embodiments, the emission parameter refers to configuration parameter information that controls the plurality of plasma generators **110** to perform an emission. In some embodiments, the plasma array controller **100** may send the emission parameter to prompt information of a terminal device to prompt a user to modify the arrangement **730** of the plurality of plasma generators **110**. In some embodiments, the plasma array controller **100** may control the plurality of plasma generators **110** based on the emission parameter to drive the plurality of plasma generators **110** to emit corresponding high-frequency jets. In some embodiments, the emission parameter may include the arrangement **730**. The arrangement **730** may include one or more of an equally spaced arrangement, an uneven arranging, a circular arrangement, or the like.

(97) In some embodiments, the plasma array controller **100** may determine the emission parameter of the plurality of plasma generators **110** based on a gas flow characteristic. Since the gas pressure characteristic sequence reflects the emission parameter, in some embodiments, in a database of the plasma array controller **100**, different arrangements **730** may correspond to different predetermined gas pressure characteristic sequence for different flow scenarios. Correspondingly, in some embodiments, the plasma array controller **100** may select, based on the distance between the gas pressure sequence **710** and the predetermined gas pressure sequence **720**, the arrangement **730** of the corresponding plasma generator **110**. For example, the plasma array controller **100** may select the arrangement **730** corresponding to the predetermined gas pressure sequence **720** with the smallest distance between the gas pressure sequence **710** and the predetermined gas pressure sequence **720**.

(98) In some embodiments of the present disclosure, determining the emission parameter sent to the terminal device based on the distance between the gas pressure sequence **710** and the predetermined gas pressure sequence **720** may prompt the user to modify the arrangement **730** of the plurality of plasma generators **110** in time, thereby quickly and efficiently achieving the required predetermined gas pressure sequence **720**.

(99) In some embodiments, as illustrated in FIG. 7, the emission parameter may further include a time delay and a phase **760** of the high-frequency jets output from the plurality of plasma generators **110**.

(100) The time delay refers to a certain time delay set between different plasma generators **110**. In some embodiments, the time delay may allow for stronger interaction of the generated high-frequency transverse waves, such as synthesis, superimposition, interference, or the like. In some embodiments, the plasma array controller **100** may control the time delay to superimpose a plurality of the high-frequency jets together in a specific manner, thereby achieving a stronger plasma flow control effect.

(101) The phase refers to a certain phase difference set between different plasma generators **110**. In some embodiments, the plasma array controller **100** may control a phase difference of the plurality of plasma array generators to adjust a propagation speed and a direction of plasma waves. For example, the phase control may be configured to adjust characteristic parameters such as an amplitude, a frequency, a phase, or the like, of the plasma waves.

(102) In some embodiments, as illustrated in FIG. 7, the plasma array controller **100** may also predict, based on a predetermined high-frequency transverse wave characteristic **740**, the time delay and the phase **760** of the high-frequency jets output from the plurality of plasma generators **110** through an emission prediction model **750**, the emission prediction model **750** being a machine learning model; and determine the arrangement **730** of the plurality of plasma generators **110** based on the time delay and the phase **760** of the high-frequency jets.

(103) In some embodiments, the emission prediction model **750** may be a machine learning model, such as a neural network model (NN), etc.

(104) In some embodiments, an input of the emission prediction model **750** may include the high-

frequency transverse wave characteristic **740**. The high-frequency transverse wave characteristic **740** refers to a characteristic parameter of required high-frequency transverse waves. In some embodiments, the high-frequency transverse wave characteristic **740** may include characteristic parameters such as a wavelength, an amplitude, a frequency, a power spectrum, or the like. In some embodiments, the high-frequency transverse wave characteristic **740** may be determined by manual input or historical transition data, etc.

(105) In some embodiments, an output of the emission prediction model **750** may include the time delay and the phase **760** of the high-frequency jets output from the plurality of plasma generators **110**. In some embodiments, the time delay and the phase **760** of the high-frequency jets may be affected by the arrangement **730** of the plurality plasma array generators. Since the arrangement **730** of the plurality plasma array generators affects a propagation path and time of the high-frequency jets, which in turn affects a magnitude and a variation of the time delay and the phase, the arrangement **730** of the plurality of plasma array generators may be further determined based on the predicted time delay and the phase.

(106) Merely by way of example, if the predicted phase difference and the predicted time delay values between different plasma generators are small, the equally spaced arrangement may be selected. If the calculated phase difference and the time delay values are large, the arrangement **730** may be adjusted to use another arrangement **730**, such as a diagonal arrangement, a symmetrical arrangement, etc.

(107) In some embodiments, the emission prediction model **750** may be trained based on a large amount of training data. The training data may include training samples and labels. For example, the training samples may include sample high-frequency transverse wave characteristics **740**, and the labels of the training samples may include time delays and phase parameters controlled by the plasma array generators corresponding to the sample high-frequency transverse wave characteristics **740**. The sample high-frequency transverse wave characteristics **740** may be historically generated high-frequency transverse wave characteristics satisfying requirements. In some embodiments, the plasma array controller **100** may obtain the high-frequency transverse wave characteristic **740** by detecting with a professional instrument, such as a spectrum analyzer, an oscilloscope, etc.

(108) In some embodiment of the present disclosure, the plasma array controller **100** may consider information such as the required high-frequency transverse wave characteristic **740** when predicting the time delay and the phase **760** of the high-frequency jets and the arrangement **730** of the plurality of plasma generators **110**, which makes the predicted time delay and the phase **760** and the arrangement **730** more realistic and improve the accuracy of transition. Meanwhile, the prediction efficiency may be improved using the machine learning model for prediction, thereby improving the timeliness of transition control.

(109) The above description is made with respect to specific embodiments of the present disclosure. Other embodiments are within the scope of the appended claims. In some embodiments, the actions or steps recorded in the claims may be performed in a different order than in the embodiments and still achieve the desired results. In addition, the processes depicted in the accompanying drawings do not necessarily require a specific or sequential order as illustrated to achieve the desired results. In some embodiments, multitasking and parallel processing are also possible or may be advantageous.

(110) In the 1930s, improvements to a technology could clearly distinguish between hardware improvements (e.g., improvements to circuit structures such as diodes, transistors, switches, etc.) and software improvements (improvements to method processes). However, as technology has evolved, many of today's method flow improvements can now be seen as direct improvements to hardware circuit structures. Designers almost always obtain the corresponding hardware circuit structure by programming the improved method flow into the hardware circuit. Therefore, it cannot be said that an improvement in a method flow cannot be realized with a hardware entity module.

For example, a programmable logic device (PLD) (e.g., a field programmable gate array (FPGA)) is one such integrated circuit whose logical function is determined by user programming of the device, which is programmed by the designer to “integrate” a digital system on a PLD, without the need to ask the chip manufacturer to design and produce a dedicated IC chip. In addition, nowadays, instead of manually producing integrated circuit chips, this programming is often done using “logic compilers” software, which is similar to the software compiler used during program development. Furthermore, the compilation of the original code before it is also done using a specific programming language, referred to as a Hardware Description Language (HDL). There are many HDLs, such as Advanced Boolean Expression Language (ABEL), Altera Hardware Description Language (AHDL), Confluence, Cornell University Programming Language (CUPL), HDCal, Java Hardware Description Language (JHDL), Lava, Lola, MyHDL, PALASM, Ruby Hardware Description Language (RHDL). Very-High-Speed Integrated Circuit Hardware Description Language (VHDL) and Verilog are the most commonly used today. It should also be clear to those skilled in the art that a hardware circuit implementing the logical method flow can easily be obtained by simply programming the method flow into an integrated circuit with a little bit of logic programming in one of the above-mentioned hardware description languages.

(111) The controller may be implemented in any suitable manner. For example, the controller may take the form of a computer chip or entity, or a processor and storage media containing computer-readable program code that can be executed by the processor, such as software or firmware, or logical gates, switches, Application Specific Integrated Circuit (ASIC), programmable logic controllers, and embedded microcontrollers. Examples of controllers include but are not limited to microcontrollers such as ARC 625D, Atmel AT91SAM, Microchip PIC18F26K20, and Silicone Labs C8051F320, where the storage controller may also be implemented as part of the control logic of the memory. Those skilled in the art also know that in addition to being implemented in a pure computer-readable program code manner, it is also possible to implement a controller that realizes the method by programming the steps of the method into a logic gate, switch, dedicated integrated circuit, programmable logic controller, and embedded microcontroller. Accordingly, such a controller can be considered a hardware component, and structures within it that are used to implement various functions can be viewed as structures within a hardware component. Or even, the devices used to implement the various functions can be considered as both a software module that implements the method and a structure within the hardware component.

(112) The systems, apparatuses, modules, or units elucidated in the above embodiments may specifically be realized by a computer chip or entity, or by a product having a certain function. One exemplary implementation device is a computer. Specifically, the computer may be, for example, a personal computer, a laptop computer, a cellular telephone, a camera phone, a smart phone, a personal digital assistant, a media player, a navigation device, an email device, a game console, a tablet computer, a wearable device, or any combination thereof.

(113) For the convenience of description, the above devices are described in terms of functions divided into various units separately. Of course, it is possible to realize the functions of the various units in the same or more than one software and/or hardware when implementing the embodiments of the present disclosure.

(114) Those skilled in the art should understand that one or more embodiments of the present disclosure may be provided as a method, system, or computer program product. Accordingly, the one or more embodiments of the present disclosure may be in the form of a fully hardware embodiment, a fully software embodiment, or an embodiment that combines software and hardware aspects. Furthermore, the present disclosure may take the form of a computer-usable storage medium (including, but not limited to, disk memory, CD-ROM, optical memory, or the like) on which one or more computer-usable program code is contained in the form of a computer program product.

(115) The present disclosure is described with reference to flowcharts and/or block diagrams of

methods, apparatus (systems), and computer program products according to the embodiments of the present disclosure. It should be understood that a combination of each of the processes and/or boxes in the flowchart and/or block diagrams, as well as the processes and/or boxes in the flowchart and/or block diagrams, may be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general-purpose computer, a special-purpose computer, an embedded processor, or other programmable data-processing device to produce a machine, such that the instructions executed by the computer or other programmable data-processing device's processor executes the instructions to produce a machine for implementing a function specified in one or more processes of a flowchart and/or one or more boxes of a block diagram.

(116) These computer program instructions may also be stored in a computer-readable memory capable of directing a computer or other programmable data processing device to operate in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture comprising an instruction device that implements a function specified in one or more processes of a flowchart and/or one or more boxes of a block diagram.

(117) These computer program instructions may also be loaded onto a computer or other programmable data-processing device, such that a series of operational steps are performed on the computer or other programmable device to produce computer-implemented processing, such that the computer or other programmable device provides steps for implementing a function specified in one or more processes of a flowchart and/or one or more boxes of a block diagram.

(118) In a typical configuration, a computing device includes one or more processors (CPUs), input/output interfaces, network interfaces, and memory.

(119) The memory may include forms of non-permanent memory, random-access memory (RAM), and/or non-volatile memory in computer-readable media, such as read-only memory (ROM) or flash memory (flash RAM). The memory is an example of computer-readable media.

(120) The computer-readable media, including permanent and non-permanent, removable and non-removable media, may be used by any method or technique to implement information storage. Information may be computer-readable instructions, data structures, program modules, or other data. Examples of storage media for computers include, but are not limited to, phase-change memory (PRAM), static random-access memory (SRAM), dynamic random-access memory (DRAM), other types of random-access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), flash memory or other memory technology, read-only CD-ROM, digital versatile discs (DVDs), or other optical storage, magnetic cartridge tapes, magnetic tape magnetic disk storage, or other magnetic storage devices, or any other non-transfer media that can be used to store information that can be accessed by a computing device. The computer-readable media, as defined herein, do not include transient computer-readable media (transitory media), such as modulated data signals and carriers.

(121) One or more embodiments of the present disclosure may be described in the general context of computer-executable instructions executed by a computer, such as a program module. In general, program modules include routines, programs, objects, components, data structures, or the like that perform particular tasks or implement particular abstract data types. One or more embodiments of the present disclosure may also be practiced in distributed computing environments in which tasks are performed by remote processing devices that are connected via a communication network. In distributed computing environments, program modules may be located in local and remote computer storage media including storage devices.

(122) Each of the embodiments in the present disclosure is described in a recursive manner, and it is sufficient to refer to each other for identical and similar portions of each embodiment, with each embodiment focusing on differences from the other embodiments. In particular, for the system embodiments, the descriptions are simpler because they are basically similar to the method embodiments, and it is sufficient to refer to the part of the method embodiments for the relevant

description.

(123) The basic concept has been described above. Obviously, for those skilled in the art, the above detailed disclosure is only an example, and does not constitute a limitation to the present disclosure. Although not expressly stated here, those skilled in the art may make various modifications, improvements and corrections to the present disclosure. Such modifications, improvements and corrections are suggested in this disclosure, so such modifications, improvements and corrections still belong to the spirit and scope of the exemplary embodiments of the present disclosure.

(124) Meanwhile, the present disclosure uses specific words to describe the embodiments of the present disclosure. For example, “one embodiment”, “an embodiment”, and/or “some embodiments” refer to a certain feature, structure or characteristic related to at least one embodiment of the present disclosure. Therefore, it should be emphasized and noted that references to “one embodiment” or “an embodiment” or “an alternative embodiment” two or more times in different places in the present disclosure do not necessarily refer to the same embodiment. In addition, certain features, structures, or characteristics in one or more embodiments of the present disclosure may be properly combined.

(125) In addition, unless clearly stated in the claims, the sequence of processing elements and sequences described in the present disclosure, the use of counts and letters, or the use of other names are not used to limit the sequence of processes and methods in the present disclosure. While the foregoing disclosure has discussed by way of various examples some embodiments of the invention that are presently believed to be useful, it should be understood that such detail is for illustrative purposes only and that the appended claims are not limited to the disclosed embodiments, but rather, the claims are intended to cover all modifications and equivalent combinations that fall within the spirit and scope of the embodiments of the present disclosure. For example, although the implementation of various components described above may be embodied in a hardware device, it may also be implemented as a software only solution, e.g., an installation on an existing server or mobile device.

(126) In the same way, it should be noted that in order to simplify the expression disclosed in this disclosure and help the understanding of one or more embodiments of the invention, in the foregoing description of the embodiments of the present disclosure, sometimes multiple features are combined into one embodiment, drawings or descriptions thereof. This method of disclosure does not, however, imply that the subject matter of the disclosure requires more features than are recited in the claims. Rather, claimed subject matter may lie in less than all features of a single foregoing disclosed embodiment.

(127) In some embodiments, counts describing the quantity of components and attributes are used. It should be understood that such counts used in the description of the embodiments use the modifiers “about”, “approximately” or “substantially” in some examples. Unless otherwise stated, “about”, “approximately” or “substantially” indicates that the stated figure allows for a variation of +20%. Accordingly, in some embodiments, the numerical parameters used in the disclosure and claims are approximations that can vary depending upon the desired characteristics of individual embodiments. In some embodiments, numerical parameters should consider the specified significant digits and adopt the general digit retention method. Although the numerical ranges and parameters used in some embodiments of the present disclosure to confirm the breadth of the range are approximations, in specific embodiments, such numerical values are set as precisely as practicable.

(128) Each of the patents, patent applications, publications of patent applications, and other material, such as articles, books, specifications, publications, documents, things, and/or the like, referenced herein is hereby incorporated herein by this reference in its entirety for all purposes, excepting any prosecution file history associated with same, any of same that is inconsistent with or in conflict with the present document, or any of same that may have a limiting affect as to the

broadest scope of the claims now or later associated with the present document. By way of example, should there be any inconsistency or conflict between the description, definition, and/or the use of a term associated with any of the incorporated material and that associated with the present document, the description, definition, and/or the use of the term in the present document shall prevail.

(129) In closing, it is to be understood that the embodiments of the application disclosed herein are illustrative of the principles of the embodiments of the application. Other modifications that may be employed may be within the scope of the application. Thus, by way of example, but not of limitation, alternative configurations of the embodiments of the application may be utilized in accordance with the teachings herein. Accordingly, embodiments of the present application are not limited to that precisely as shown and described.

Claims

1. A transverse wave excitation plasma array generator, comprising a plasma array controller and a controlled profile, wherein the plasma array controller includes a plurality of plasma generators, the plasma array controller refers to an electronic device that performs transition control, the plasma array controller is mounted on the controlled profile, and a transition refers to a process of a fluid transitioning from a laminar flow state to a turbulent flow state; the plasma array controller is configured to: control the plurality of plasma generators to inject an energy into atoms and molecules and convert the atoms and the molecules into ions to generate high-frequency jets under an excitation of high-frequency electricity, wherein a frequency of the high-frequency electricity is within a range of 1 kHz-3 kHz, a voltage of the high-frequency electricity is within a range of 1 kV-10 kV, and a frequency of the high-frequency jets is within a range of 1 kHz-3 kHz, obtain spatially distributed high-frequency transverse waves, and generate high-frequency excitation for controlling a second mode of a boundary layer of the controlled profile to promote the transition of the boundary layer of the controlled profile; wherein the plurality of the plasma generators are connected with a capacitor and an inductor to form an inductance-capacitance resonance circuit, the inductance- capacitance resonance circuit is set at a predetermined loop position, each of the plurality of plasma generators includes a cavity and electrodes disposed at two sides of the cavity, and the electrodes generate, based on the high-frequency electricity, the high-frequency jets; and the plasma array controller is configured as that the each of the plurality of plasma generators is connected in series with a plurality of capacitor units and a plurality of inductive coils to form a multi-level array loop; an input end of a lower array loop is connected with a downstream of an inductive coil of a higher array loop to form a loop module; and each loop module is connected in series with the next loop module to form a loop module group, and each loop module group is connected in parallel with the next loop module group to form the plasma array controller.

2. The transverse wave excitation plasma array generator of claim 1, wherein a material of the cavity includes a ceramic, and a material of the electrodes includes tungsten.

3. The transverse wave excitation plasma array generator of claim 1, wherein a ratio of a diameter of each of the electrodes to an inner diameter of the cavity is within a range of 0.05-0.08.

4. The transverse wave excitation plasma array generator of claim 1, wherein a ratio of a spacing of the plurality of plasma generators in an X-direction to a spacing of the plurality of plasma generators in a Z-direction is within a range of 0.8-1.2.

5. The transverse wave excitation plasma array generator of claim 4, wherein a ratio of a count of the plurality of plasma generators distributed in the X-direction to a count of the plurality of plasma generators distributed in the Z-direction is within a range of 0.8-1.2.

6. The transverse wave excitation plasma array generator of claim 1, wherein output characteristics of the plurality of plasma generators satisfy a predetermined output relationship expressed as:

$Y=F(\omega,t,\tau,\varphi)$ wherein Y denotes the output characteristics of the plurality of plasma generators, ω

denotes a jet frequency of the plurality of plasma generators, t denotes a time sequence of the plurality of plasma generators generating the high-frequency jets, τ denotes a spatial distribution delay matrix of the transverse wave excitation plasma array generator, and ϕ denotes a spatial distribution phase matrix of the transverse wave excitation plasma array generator.

7. The transverse wave excitation plasma array generator of claim 1, wherein one or more gas pressure sensors are provided within a predetermined range where the boundary layer of the controlled profile is located, and the plasma array controller is further configured to: obtain a gas pressure sequence detected by the one or more gas pressure sensors, and determine, based on a distance between the gas pressure sequence and a predetermined gas pressure sequence, an emission parameter of the plurality of plasma generators, the emission parameter including an arrangement of the plurality of plasma generators.

8. The transverse wave excitation plasma array generator of claim 7, wherein the emission parameter further includes a time delay and a phase of the high-frequency jets output from the plurality of plasma generators, and the plasma array controller is further configured to: predict, based on a predetermined high-frequency transverse wave characteristic, the time delay and the phase of the high-frequency jets output from the plurality of plasma generators through an emission prediction model, the emission prediction model being a machine learning model; and determine the arrangement of the plurality of plasma generators based on the time delay and the phase of the high-frequency jets.

9. The transverse wave excitation plasma array generator of claim 7, wherein a count and an arrangement of the one or more gas pressure sensors are related to a type of transition and a demand degree of transition of the boundary layer.

10. The transverse wave excitation plasma array generator of claim 8, wherein the time delay refers to a certain time delay set between the plurality of plasma generators, and the plasma array controller is configured to control the time delay to superimpose a plurality of the high-frequency jets together in a specific manner.

11. The transverse wave excitation plasma array generator of claim 9, wherein the plasma array controller is further configured to: in response to determining that the demand degree of transition is low, provide a relatively small count of the gas pressure sensors near the boundary layer.

12. The transverse wave excitation plasma array generator of claim 1, wherein the second mode refers to an inherent vibration characteristic of the boundary layer with a predetermined inherent frequency, the predetermined inherent frequency is determined based on a parameter and a morphology of the boundary layer, and the boundary layer refers to a fluid boundary layer that attaches to surface of a solid when the fluid flows along the surface of the solid or when the solid moves in the fluid.

13. The transverse wave excitation plasma array generator of claim 1, wherein in the series connection of a plurality of loop modules, an input end of an inductive coil in a first array loop of the next loop module is connected with an output end of a plasma generator of a previous loop module to form the loop module group.

14. The transverse wave excitation plasma array generator of claim 1, wherein in the parallel connection of a plurality of loop module groups, input ends of a plurality of inductive coils in a first array loop of each of the plurality of loop module groups are connected, and output ends of a plurality of plasma generators in a last array loop of the each of the plurality of loop module groups are connected, and the connected input ends and the connected output ends receive the excitation of the high-frequency electricity, respectively, to form the plasma array controller.

15. The transverse wave excitation plasma array generator of claim 1, wherein the cavity is provided with through holes to allow the ions to be ejected from the cavity to generate the high-frequency jets.

16. The transverse wave excitation plasma array generator of claim 1, wherein the predetermined loop position refers to a position of a loop under the excitation by the high-frequency electricity,

and when a large number of plasma generators are required, the plurality of the plasma generators are set to be separately located on a loop array.
